

Genome-wide analysis of the beta-glucosidase gene family in maize (*Zea mays* L. var B73)

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Abstract The hydrolysis of beta-D-glucosidic bonds which is required for the liberation of many physiologically important compounds is catalyzed by the enzyme beta-glucosidase (BGLU, EC 3.2.1.21). BGLUs are implicated in several processes in plants, such as the timely response to biotic and abiotic stresses through activation of phytohormones and defense compounds. We identified 26 BGLU isozymes in the genome of the maize inbred B73 and propose a standardized nomenclature for all *Zea mays* BGLU paralogs (*Zmbglu1*–*Zmbglu26*). We characterized their intron–exon structure, protein features, phylogenetic relationships, and measured their expression and activity in various tissues under different environmental conditions.

Sequence alignments revealed some characteristic motifs (conserved amino acids) and specific differences among different isozymes. Analysis of putative signal peptides suggested that some BGLUs are plastidic, whereas others are mitochondrial, cytosolic, vacuolar or secreted. Microarray and RT–PCR analysis showed that each member of the *Zmbglu* family had a characteristic expression pattern with regard to tissue specificity and response to different abiotic conditions. The source of variance for gene expression was highest for the type of organ analyzed (tissue variance) than for the growth conditions (environmental variance) or genotype (genetic variance). Analysis of promoter sequences revealed that each *Zmbglu* paralog possesses a distinct set of *cis* elements and transcription factor binding sites. Since there are no two *Zmbglu* paralogs that have identical molecular properties, we conclude that gene subfunctionalization in maize occurs much more rapidly than gene duplication.

This article is dedicated to the memory of the late Roberto Carlos Vázquez Juárez and Felix Victor Córdoba Alva.

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Introduction

Many chemical compounds in the vegetable kingdom are present as soluble and inactive conjugates of glucose. For example, one such naturally occurring glucoside is 2-O- β -D-glucopyranosyl-4-hydroxy-7-methoxy-1,4-benzoxazin-3-one (DIMBOA-Glc). Other examples of such compounds are the indoxyl-beta-glucosides and zeatin-9-glucoside belonging to the auxin and cytokinin family of hormones. The enzyme β -glucosidase (BGLU; EC 3.2.1.21) catalyses the hydrolysis of β -D-glucosidic bonds, releasing glucose

and activating such compounds (Babcock and Esen 1994). In plant cells, this activity is important for the activation of defense chemicals and also for the release of active hormones (Brzobohaty et al. 1993). The maize enzyme that can be extracted from young coleoptiles hydrolyzes DIM-BOA-Glc and other aryl or alkyl β -D-glucosides (Esen 1992; Babcock and Esen 1994; Czjzek et al. 2001). It can also hydrolyse auxin-glucosides and other hormones (Campos et al. 1992; Campos et al. 1993; Feldwisch et al. 1994). The β -glucosidase enzyme (BGLU) is often compartmentalized at the subcellular level (either in the plastid, cytosol, vacuole or apoplast) and can be found mainly in young vegetative parts (Kristoffersen et al. 2000; Biely et al. 2003).

It had been previously reported that *Zea mays* has only two β -glucosidase (*bglu*) isogenes, namely *glu1* and *glu2* (Yu et al. 2009a) (here renamed to *bglu1* and *bglu2*). The two isozymes are encoded by two different genes that have a ~90% sequence identity at the nucleotide level. The whole of the maize β -glucosidase activity was formerly believed to be represented by those isozymes, BGLU1 and BGLU2 (Esen 1992; Ebisui et al. 1998; Czjzek et al. 2001). In the MaizeGDB database, *glu2* has been previously mapped to chromosome 2 (www.maizegdb.org). However the MaizeSequence database maps *glu2* on chromosome 10. The ambiguity of the mapping data in the MaizeGDB database might reflect cross-hybridization of molecular markers due to the presence of more than 2 genes in the maize genome. It is typical of plant enzymes to be present as multigene families which are expressed differentially. For example, in the *Arabidopsis* genome, the glycoside hydrolase family 1 consists of 47 genes (Xu et al. 2004) and most members have distinct gene expression pattern (see Genevestigator database).

We have performed a genome-wide BGLU protein survey using the recently completed B73 maize genome (Schnable et al. 2009). A comparative analysis allowed us to classify the genes into 4 distinct groups. We propose a standardized nomenclature for all 26 beta-glucosidase paralogs in maize (*Zmbglu1*–*Zmbglu26*). This naming convention will help researchers and breeders to use a uniform numbering for all *bglu* loci in future experiments or projects. We also analyzed the expression pattern of the *bglu* family using Microarrays and RT–PCR. We wanted to estimate the degree of variability of gene expression on the basis of the growing location (LOC), environmental condition such as drought stress (ENV), genotype (GEN) and tissue (TIS). In addition, the promoter regions of the *bglu* genes were analyzed in order to identify transcription binding sites and regulatory sequences that could modulate the expression in dependence from the factorial combination of LOC $\times$ ENV $\times$ GEN $\times$ TIS.

Materials and methods

Source of data and employed tools for the genome-wide analysis

Recently, a draft sequence of the maize genome (*Zea mays* var B73) has been released by the B73 maize sequence consortium (Schnable et al. 2009). Using the data that can be downloaded at www.maizesequence.org (release version 4a.53 and release 5b.60), a detailed bioinformatic analysis of all the β -glucosidase genes and proteins was made as follows. Genes were identified by iterative protein BLAST searches (Altschul et al. 1997). Sequence alignments and dendograms were done using the program CLC Combined Workbench version 3.5.1 (www.clcbio.com). Some alignments were also done with Clustal (Larkin et al. 2007). Full length protein sequences were used to classify the genes into subgroups. In order to predict the targeting of the enzymes into different subcellular locations, the proteins were analyzed for the presence or absence of signal peptides using the TargetP v1.1 prediction tool (www.cbs.dtu.dk/services/TargetP/). Subcellular localization was also predicted with Predotar v1.03 (www.genoplante-info.infobiogen.fr/predotar/) and WoLF Psort (wolfsort.org). The molecular weight for each protein was calculated using ProtParam from the ExPASy Proteomics Service (us.expasy.org/tools/protparam.html) (Gasteiger et al. 2005). The theoretical isoelectric point (pI) of the full length proteins was calculated with ProtParam (Gasteiger et al. 2005), with the EMBL pI tool (<http://www3.embl.de/cgi/pi-wrapper.pl>) which is based on (Lehninger 1979), and with Sequence Manipulation Suite (Stothard 2000) which gives the same result as EMBOSS pepstats. Putative N-Glycosylation sites were predicted using the NetNGlyc 1.0 Server (www.cbs.dtu.dk/services/NetNGlyc/).

The gene promoter regions of all *Zmbglu* genes were obtained by selecting an approximate of 570 nucleotides upstream of the predicted start sites. The protein start sites were found by searching the open reading frames (ORFs) and by aligning the genomic sequences to the coding determining sequences (CDS) as available at the maizesequence database. The gene promoter regions were analyzed with the Promoter Scan tool version 1.7 (www.bimas.cit.nih.gov/molbio/proscan/) which is based in sequence similarity to the binding sites of eukaryotic polymerase II. The transcription factor binding sites were found using the Tfsitescan tool (www.ifti.org/cgi-bin/ifti/Tfsitescan.pl) (Prestridge 1995) and the *cis* elements were found using the PLACE database (www.dna.affrc.go.jp/PLACE/signalscan.html) (Higo et al. 1998; Higo et al. 1999).

Plant material and culture conditions

The experiments were done with different maize genotypes that were grown or obtained as follows: One set of plants were grown in the open field in México (Tlaltizapan trials 2004) and another set was grown in different controlled environments (greenhouse or growth chambers). We used four maize lines from the CIMMYT C5 mapping population: P1 (Ac7643) and P2 (Ac7729/TZSRW), respectively the short anthesis-silking interval (ASI) parent and the long ASI parent, and RIL 151 and RIL 245, contrasting lines, respectively very drought tolerant and very susceptible. Plants were grown under two different environmental conditions: well watered [(WW) water potential of -0.1 MPa] and drought stress (SS) during the vegetative stage (-0.5 MPa at sampling in the stressed plants). Approximately 30 individual plants were dissected and different tissues (silks, leaves and roots) were harvested. The 6-cm basal parts of the youngest growing leaves (in which cells divide and/or expand) were sampled and pooled. Samples of hidden parts of the silks were sampled at predawn leaf water potentials of -0.1 and -0.5 MPa (WW and SS, respectively) in field and greenhouse. Samples of roots (7–10 mm apical parts of the primary and secondary roots in which cells divide and expand) were collected in the same set of lines at the two studied water potential in the growth chamber at ETH, Switzerland. In addition to the silk samples harvested in field trials in Tlaltizapan (silk_u_WW and silk_u_SS) we also harvested silks samples from the CIMMYT C5 lines grown in a greenhouse in Switzerland under SS and WW conditions (silk_WW and silk_SS).

We also grew B73 plants in the greenhouse in Irapuato (México) and exposed mature green leaves to two different conditions for 3 h before harvesting: Light (leafL) and Dark (leafD). In addition, we grew a commercial hybrid in México and harvested embryo from kernels that were pollinated with white parent (embryoW) or a yellow parent (embryoY).

For the salt stress and mechanical damage assays, a pool of 200 seeds from the maize inbred B73 were used. Seeds were surface sterilized by soaking them in 10% (v/v) hypochlorite solution (commercial household bleach) for 5 min, and then rinsed with distilled water. Germination pans were prepared with sterile wet tissues and kept on a humid chamber at 24°C in the dark during the experiment. Abiotic stress was induced by soaking seedlings 5 days after germination with 200 ml of 300 mM NaCl. For mechanical damage stress experiments, 5-day seedlings were cut around the coleoptiles to simulate the mechanical damage induced by insects or other pests. Samples were taken at given timepoints (see figure legends) after salt or mechanical treatment (pool of 5 coleoptiles each).

Carefully harvested samples were immediately frozen in liquid N_2 and stored at -80°C until analysis.

Microarrays

All microarray experiments were performed using the 46 K chip from the University of Arizona. The chip is a custom made oligonucleotide array containing 46,128 70-mer oligos printed on a single glass slide. We performed 30 different hybridization experiments. In addition, we also included publicly available data from different hybridization experiments published in the GeneOminbus section of the NCBI database (GEO:GSE10543 and GEO:GSE10449). The samples we additionally included were 15 day whole plantlets of different genotypes (B73, Mo17, W22, Oh43, etc.) that were grown in USA (Stupar et al. 2008). For each genotype we first averaged the results from all replicas, and then used the mean values for further analysis. We also included endosperm samples from opaque2 and normal maize (Endosperm1 and Endosperm2, see Fig. 5) (Holding et al. 2008).

Most of our microarray expression analysis was done of SS samples in comparison to WW samples. Other treatments were light (L) and dark (D) conditions for leaves, or white (W) or yellow (Y) colored embryos (Fig. 5). We averaged the results from the 4 different CIMMYT C5 lines in order to reduce the variance of the expression data for each individual *Zmbglu* gene.

For all previously described samples, total RNA was isolated by the TRIzol method (Invitrogen). RNA was cleaned with DNase I (Invitrogen) and also additionally purified with a commercial tube kit for RNA isolation (Invitrogen). RNA integrity was confirmed by 2% agarose-formaldehyde gel electrophoresis (Sambrook et al. 1989).

Expression analysis of the *Zmbglu* gene family using the oligoarray data

In order to identify all the corresponding probes in the maize array (University of Arizona) we blasted all available 70mer oligos against all gene accessions of the maize genome (Schnable et al. 2009). We found that the correspondence table is ambiguous in both ways (Fig. S2). This means that the same gene (GRMZM entry) can have matches with several oligoprobes (MZ entry), whereas, the same MZ oligo can also have matches with different GRMZM genes of the same family. For example the gene *bglu1* (=GRMZM2G016890) had significant matches with up to 10 different oligo probes in the maizearray (Fig. S2). The alignments had different identity values in the Blast comparison to *bglu1* (for example 100% MZ00057235, 97% MZ00043439, 83% MZ0004348 and so forth). On the other hand, the same oligoprobe MZ00057235 in addition

to be a perfect match to *bglu1* (100% identity and bitscore of 139), it had also matches to other transcripts like (96% *bglu3* = GRMZM2G120962_T01, 96% *bglu2* = GRMZM2G008247_T01, 85% *bglu7* = GRMZM2G426467_T01 and so forth). A visualization of the bitscores of all pair comparisons are shown in the supplemental Fig. S2. Having determined the identity matrix (Fig. S2), we then applied a statistical algorithm based on the least square method in order to deconvolute the microarray data from values based on MZ probes to values based on GRMZM genes. For this we used the Solver function of Excel to find the minimum of the square sums. The above method allowed us to discriminate the expression levels of individual *bglu* genes even when there had been cross hybridization between them. The following procedure helped also to reduce the variance of the data (typically, maizearray spots have a coefficient of variation that can be as high as 30–50%). In addition of averaging values among spots within the same slide, we also averaged values across slides (individual slides were previously normalized for a mean spot intensity of 1000). In order to increase reliability, we also averaged the data across all CIMMYT C5 maize genotypes. For example, the data for leaf tissue shown in Fig. 5 was the average of 12 independent samples each for SS or WW.

Primer design for RT-PCR

The primers for ubiquitin (GenBank accession S94464) were: ZmUBIF2 (5'-CACCTTCACCTGTTCTTCC-3') and ZmUBIR1 (5'-CCTCCAGGGTTATGGTTTTTCC-3'), for which the product size was 88 bp. All *Zmbglu* genes were aligned and low homology regions of the 3' UTR were selected to design the *Zmbglu1* and *Zmbglu2* specific primers. Sequence alignments were done using either Clustal W (Thompson et al. 1994) or CLC combined workbench (www.clcbio.com). The primer design was done using PRIMER3 (frodo.wi.mit.edu) and DNA calculator (www.sigma-genosys.com/calc/DNACalc.asp). The *Zmbglu1* and *Zmbglu2* specific primers were: *Zmbglu1F* (5'-TTAGGAGCCACCTAGTAG-3'), *Zmbglu1R* (5'-CGTGTAGTAAAGCTAAGGTTAC-3') and *Zmbglu2F3* (5'-TACTGCCTAAGGGAACGGTC-3'), *Zmbglu2R3* (5'-TCTTCTAGTGCTTGAGGG-3') respectively. They were evaluated and validated using specific clones of *Zmbglu1* and *Zmbglu2* as templates (Fig. S2).

The steady state mRNAs levels were quantified by reverse transcription and PCR (RT-PCR). The cDNA was synthesized using 5 µg of total RNA, IMPROM II reverse transcriptase (Promega) and oligo dT(12–18) in a 22 µl final reaction volume. The PCR amplification was performed in a 10 µl reaction mixture containing 0.8 µl of cDNA, 1 µl of 10× polymerase buffer, 0.6 µl of 50 mM

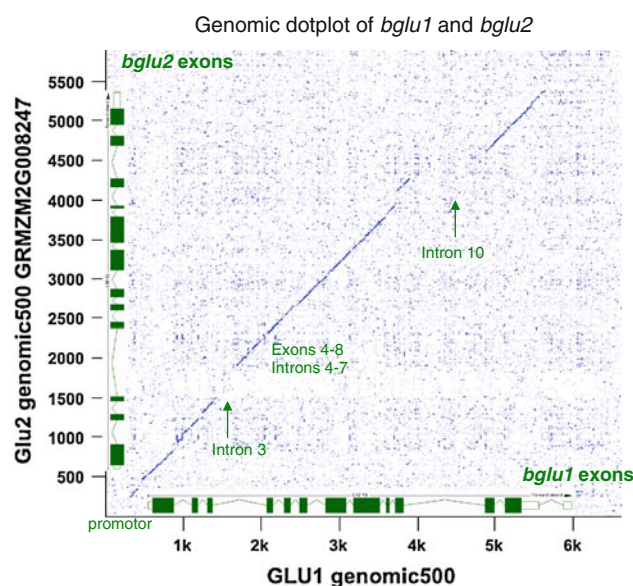


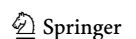
Fig. 1 Dotplot comparison of the genomic sequences of *bglu1* and *bglu2*. The intron–exon structure of each gene is shown in green on both axes. Sequence similarity is shown as blue dots in the graph. Sequence colinearity is revealed as a diagonal blue line. As it can be seen, the regions of sequence similarity are not restricted to exons, but are also present within introns. For example, introns 4–7 show a strong similarity between *bglu1* and *bglu2*. Exceptions are intron 3 and intron 10 that are dissimilar. For example, the gap of intron 10 can represent a genomic insertion (in *bglu1*) or deletion (in *bglu2*). Sequence similarity stretches also to parts of the promoter, indicating some common *cis*-acting elements for transcriptional regulation of the two genes

MgCl₂, 0.52 µl of 10 mM dNTPs, 0.1 µM of each primer and 0.4 U of Taq polymerase (Invitrogen). The PCR amplification conditions were: 3 min at 94°C followed by 27 cycles of 1 min at 94°C, 1 min at 59°C and finally 30 s at 72°C, followed by 72°C for 5 min. The number of PCR cycles was further evaluated to reach the exponential-linear phase. PCR amplicons were analyzed by electrophoresis in 1.5% (w/v) agarose run at 70 V/20 cm for 60 min. Signal quantification was performed with the Doc-It® LS Image Analysis Software. Values were normalized with ubiquitin (house-keeping gene).

Tissue extraction for activity measurements

The β-glucosidase activity was evaluated in coleoptile homogenates (7 post-germination days) incubated with

Fig. 2 Global alignment of all β-glucosidase proteins. The parameters employed for alignment were: gap cost 40 and gap extension 10 and gap end as any other. The different colors highlight the chemical properties of residues (rasmol color coding: negatively charged (red), positively charged (dark blue), neutral (light blue), hydrophobic (green), sulphur containing (yellow), etc.). The sequence conservation is shown as a percentage bar-score below. At the very bottom, the sequence logo is shown, summarizing the occurrence of given amino acids at specific positions



300 mM NaCl for different hours under saline stress. Twenty four seeds (maize inbred B73) were used each day. Extracts were prepared according to (Martinez-Cruz et al. 2001). Briefly, around 20 germinated seedlings were separated in the different tissues (seeds, roots, coleoptiles, leaves and stems) and pooled each day to be extracted with citrate/citric acid buffer (0.1 M, pH 6.4) using 10 ml g⁻¹ of biomass. Tissues were processed with a biohomogenizer (Biospec Products, Bartlesville, Ok) with 3 pulses of 1 min each on ice. Crude extract were kept at 4°C and 1 g of activated charcoal was added to remove phenols, before filtering through paper (Whatman No. 541), collecting a transparent filtrate. Activity was measured directly, or in some cases enzymes were further purified as follows. To each extract, two volumes of cold acetone (-20°C) were added drop wise, with stirring on an ice bath and then, a precipitate was formed by centrifugation at 25,000 g at 4°C for 15 minutes. Each pellet was dried with gentle air flow and then dissolved in citrate/citric acid buffer (0.1 M, pH 6.4) in a volume equal to the original biomass weight (w:v).

β -glucosidase activity

β -glucosidase activity was assayed using p -nitrophenyl- β -glucopyranoside (Sigma) as substrate (Esen and Stetler 1992). Nitrophenol was measured spectrophotometrically at 415 nm in a microplate reader (BIO-RAD, Hercules, Ca) in a total volume of 210 μ l. Extracts were diluted 1:1 with citrate/phosphate buffer (10 mM sodium citrate; 20 mM sodium phosphate; pH 5.5). Measurements were made by applying on each well 70 μ l of each extract, followed by 70 μ l of substrate (5 mM paranitrophenyl-glucopyranoside on the same buffer) and after 5 minutes, 70 μ l of 400 mM Na₂CO₃ to stop the reaction. One unit of enzyme activity was equal to 1 μ mol of nitrophenol/hour at 25°C.

Statistical procedures

For all comparisons between samples, analysis of variance (ANOVA) or Linear Modelling (LM) were done with a threshold of $P \leq 0.05$ for statistical significance. To determine significant differences in gene expression at different time-points, the data was not treated as a time-series but as an ANOVA of one factor with replicates. When significant statistical values were found for a given ANOVA factor ($P \leq 0.05$), the Tukey HSD test for comparison of means was then applied.

Statistical analysis was done mainly with the basic package of R version 2.10 (R Development Core Team 2010). Analysis of variance (aov), Linear models (lm), Principal component analysis (pca), hierarchical clustering (hclust) and heatmap biclusters were done with R using the

bioconductor, limma, lattice and pcaMethods libraries with default settings for the *aov*, *lm*, *pca*, *histogram*, *hclust*, *dist*, *pairs*, *heatmap* and *plot* functions (R Development Core Team 2010).

Results

Gene nomenclature and map positions of *Zmbglu1* and *Zmbglu2* in the B73 genome

For the previously defined maize genes *glu1* and *glu2* we examined the data across different databases, in order to do a careful cross-check of the published map, gene and protein information. Taking the gene naming convention that has been applied for the β -glucosidase genes in *Ara-bidopsis* and *Oryza* (Xu et al. 2004; Opassiri et al. 2006), we decided to rename the previously characterized *glu* genes to *bglu* or *Zmbglu* (*Zea mays bglu*). The maize *bglu1* locus corresponds to the following entries in Genbank (ZMU44773, U44773, Q41761, NM_001111984) and MaizeSequence database (GRMZM2G016890). It is located in chromosome 10 at the physical coordinates 34,232,717–34,238,135 bp which correspond to the genetic coordinates 45 cM in the Genetic 2008 map and 199 cM in the IBM2 map (the difference of centimorgan units is due to the increased recombination frequency accumulated through several generations of sexual recombination in the IBM2 mapping population (Lee et al. 2002). The *Zmbglu1* gene has a length of about 5 kb (4939 bp) comprising 12 exons and 11 introns (see Accession U44773) (Esen and Bandaranayake 1998).

The *Zmbglu2* gene has a ~90% sequence identity to *Zmbglu1*, based on amino acids or nucleotides. According to MaizeGDB, the locus *glu2* is on chromosome 2, at the genetic coordinates 85 cM in the Genetic 2008 map which corresponds to bin 2.04 (<http://www.maizegdb.org/cgi-bin/displaylocusrecord.cgi?id=61768>). We found that the mapping information about the locus *glu2* on the MaizeGDB database is incorrect. *Zmbglu2* is rather located on chromosome 10, very close to *Zmbglu1* (Fig. S1). The high similarity between different *Zmbglu* genes could have caused the ambiguous mapping results in the MaizeGDB database. The possibility of gene confusion was studied by making nucleotide comparisons using the dotplot technique. When compared at the genomic sequence level, it was found that the genes not only show similarity within the coding exons, but also among the non-coding introns and UTRs (Fig. 1). This is indicative that the gene duplication occurred at the genomic level after the intron and exon structure of the *bglu* precursor gene was defined in a common ancestor. Since the gene duplication of *bglu1* and *bglu2* there has been high sequence divergence due to

transposons. Not all intronic sequences are conserved, with the differences being particularly noticeable at intron 3 and intron 10 (Fig. 1). The gap in the dotplot at intron 10 (Fig. 1) can represent a DNA insertion (in *bglu1*) or deletion (in *bglu2*). In summary, the maize BGLU2 protein corresponds to following entries in Genbank (ZMU44087, U44087, NM_001112422), and in MaizeSequence database (GRMZM2G008247). The gene is located on chromosome 10 at the physical coordinates 37,754,967–37,759,771 bp corresponds to bin 10.03 at the approximate genetic coordinate 195 cM in the IBM2 map.

Identification of all beta-glucosyl hydrolase-like genes in the maize genome

Considering the previously mentioned discrepancy in the MaizeGDB database, we hypothesized that several β -glucosidase sequences that are duplicated across the maize chromosomes have lead to misleading mapping results, particularly when inadequate primers had been used as markers. Therefore, we decided to localize all BGLU proteins in maize in order to make a comprehensive catalogue with a standard *bglu* gene nomenclature. We first studied the similarities and differences at the protein and DNA level of the known *bglu* genes in maize, arabidopsis and rice (Esen 1992; Xu et al. 2004; Opassiri et al. 2006). Almost all BGLU enzymes contain the same PfamB domain (PB027112) in the N-terminus followed by a PfamA domain (PF00232, glyco_hydro_1). They also contain the interpro domain (IPR001360). The proteins belong to the glycoside-hydrolase family 1, which is part of the superfamily of tim barrel glycosyl hydrolases (Clan: Glyco_hydro_tim CL0058) which contains 27 other gene families (pfam.sanger.ac.uk). In order to have an accurate figure on the number of *bglu* genes present in maize, iterative BLAST searches were made, followed by manual curation of the entries. We first found that the B73 maize genome harbors many genes that contain the Pfam domain PF00232 (Fig. S1A). We also blasted the full protein sequence of BGLU1 (=GRMZM2G016890_P01) and BGLU2 (=GRMZM2G008247_P01) to the whole maize genome and found that there were at least 40 protein entries which had highly significant matches (data not shown). The most significant matches were located on chromosomes 10 and 3 (red triangles on Fig. S1B). On chromosome 10, there were at least 4 hits, one of which corresponded to *bglu1* itself and another hit that is nearby which corresponded to *bglu2* (Fig. S1C). There were also several hits on chromosome 2 with a similarity of ~50% which could explain why MaizeGDB erroneously mapped *glu2* at chromosome 2. Combining all BLAST results, we detected more than 50 proteins in the B73 maize genome release 5b.60 that had significant similarity to BGLU proteins from

other plant species (data not shown). The original list was curated manually, eliminating protein splice variants, BGLU pseudoproteins, peptidase-like proteins, phosphodiesterase-like proteins and other similar hydrolytic enzymes. The list was reduced to a number of 26 unique *Zmbglu* genes (Table 1) that could have potential BGLU activity.

Global alignment and dendrogram BGLU proteins in the maize genome

Starting with the information from genomic release 4a.50, we started to name maize *bglu* genes consecutively according to protein similarity. When the release 5b.60 became available, new genes were added to the preliminary list of BGLU paralogs. Since the MaizeSequence database is being constantly updated, we decided to conserve the names of the *bglu* genes and number additional putative paralogs consecutively as they were identified and characterized.

We finally selected 26 proteins in order to compare them to each other by making a global alignment of their corresponding full length proteins (Fig. 2). The similarity results of the protein alignments were then used to construct a dendrogram (Fig. 3). Based on the previously mentioned strategy, we propose a standardized nomenclature of the maize *bglu* gene family from *Zmbglu1* to *Zmbglu26* classified into four separate groups (Fig. 3; Table 1).

The sequence features of the putative BGLU proteins were then compared. It was found that most maize BGLU proteins shared some peptide motifs that are typical for this enzyme family in other plant species. Highly conserved sequences were identified in the *ZmBGLU* gene family: SAYQXEG and AYRFSISWSR at the N terminus, CFXXFGDRVKXWXTFNEP, PGRCS, GNSXXEPYI-VAHN, LLAHA GWFXXP, GDYP and RLPXF in the middle and IXENG and GANVXGYFXWSLLDNFE at the C terminus of the proteins (Fig. 2). The conserved motifs were distributed along the entire TIM-barrel structure of the BGLU protein, involving both α -helices and β -strands (data not shown). Some of the conserved motifs, for example TFNEP and ENG include key amino acids involved in enzymatic catalysis (Czjzek et al. 2000; Zouhar et al. 2001). These glutamate residues were indeed located in inside the TIM-Barrel, where they interact with the substrate (data not shown. See also pdb file 1e4n).

Twenty-four of the BGLU's had all key residues involved in catalysis. However, we found that BGLU9 and BGLU10 are short proteins that do not form a full TIM-Barrel (data not shown). Both also lack the highly conserved ENG motif required for catalysis (Fig. 2). With only those exceptions, all the other BGLU proteins listed in

Table 1 Properties of BGLU isoforms in maize

Group	Gene Name	Molecular weight in Daltons (number of residues)	Theoretical pI ProtParam EMBL SMS	Prediction of location	Predicted N-glycosylation sites (position amino acid)
1	ZmBGLU1 GRMZM2G016890	64219.2 Da (566 aa)	6.23 6.22 6.68	Chloroplast ^{a,b,c}	NESF (27), NLSF (48), NCTR (537)
	ZmBGLU2 GRMZM2G008247	64120.5 Da (563 aa)	6.75 6.80 7.25	Chloroplast ^{a,b,c}	NESF (25), NLSF (45), NYTR (534)
	ZmBGLU3 GRMZM2G120962	63301.0 Da (556 aa)	6.08 6.07 6.51	Chloroplast ^{a,b,c}	NLSF (43), NGSQ (55), NGTL (154)
	ZmBGLU4 GRMZM2G014844	63435.6 Da (557 aa)	7.64 7.73 7.93	Chloroplast ^{a,b,c}	NHTA (10)
	ZmBGLU5 GRMZM2G077015	64450.7 Da (570 aa)	8.70 8.60 8.71	Chloroplast ^{a,c} Mitochondrial ^b	No sites predicted
	ZmBGLU6 GRMZM2G076946	64274.4 Da (567 aa)	8.01 7.95 8.11	Chloroplast ^{a,b,c}	NPTG (266)
	ZmBGLU7 GRMZM2G426467	58981.2 Da (515 aa)	5.08 4.90 4.88	Other ^{a,c} , none ^b	No sites predicted
	ZmBGLU8 GRMZM2G174699	34546.7 Da (509 aa)	5.81 5.71 6.09	Chloroplast ^{a,b,c}	NLSF (41), NGSQ (56), NGTL (156)
	ZmBGLU10 GRMZM2G362362	44094.8 Da (395 aa)	7.63 7.67 7.86	Mitochondrial ^a Secretory pathway ^b Other ^c	NGSL (132)

Table 1 continued

Group	Gene Name	Molecular weight in Daltons(number of residues)	Theoretical pI ProtParam EMBL SMS	Prediction of location	Predicted N-glycosylation sites (position amino acid)
2	ZmBGLU11	59917.8 Da (532 aa)	5.27	Mitochondrial ^{a,c}	NFTR (36), NFTH (73), NCTA (230), NSSV (236)
	GRMZM2G055699		5.12	Secretory pathway ^b	
	ZmBGLU12	56923.8 Da (508 aa)	5.19		
	GRMZM2G069024		4.90	Secretory pathway ^{a,b}	NSSV (222), NSTA (271), NGS (410)
			4.71		
	ZmBGLU13	59865.5 Da (531 aa)	4.68	Chloroplast ^c	
	GRMZM2G376416		6.28	Mitochondrial ^a	NSTT (234), NFTN (372), NRTR (492)
			6.30	Secretory pathway ^{b,c}	
			6.76		
	ZmBGLU14	59439.8 Da (533 aa)	5.55	Secretory pathway ^{a,b}	NATG (89), NSTT (239)
3	GRMZM2G108133		5.46		
	ZmBGLU15	59846.2 Da (539 aa)	5.73	Chloroplast ^c	
	AC234160.1_FG003		8.99	Mitochondrial ^{a,b}	NASR (3), NATA (104), NVT (455)NGTG (144), NATA (374), NPTM (440),
			9.06	Chloroplast ^c	
			9.14		
	ZmBGLU16	58437.1 Da (512 aa)	6.68	Secretory pathway ^{a,b}	NATA (90), NSTT (236), NPTM (412)
	GRMZM2G148176		6.71	Chloroplast ^c	
			7.18		
	ZmBGLU17	56804.1 Da (506 aa)	8.83	Secretory pathway ^{a,b}	NQTA (83), NSTE (278), NPTV (405), NLTR (420)
	AC217401.3_FG001		8.86		
	ZmBGLU18	56415.7 Da (502 aa)	8.95	Vacuole ^c	
	GRMZM2G118003		8.91	Secretory pathway ^{a,b}	NPTI (404), NLTR (419)
			8.97		
	ZmBGLU19	57859.7 Da (502 aa)	9.05	Peroxisome ^c	
	GRMZM2G112704		8.32	Secretory pathway ^{a,b}	NATA (83), N FTF (320), NATE (351), NEIL (418)
			8.25		
	ZmBGLU20	58213.6 Da (511 aa)	8.36	Chloroplast ^c	
	GRMZM2G163544		5.84	Secretory pathway ^{a,b,c}	NGTG (107), NLTR (478)
			5.81		
	ZmBGLU21	53450.3 Da (468 aa)	6.23	Mitochondrial ^{a,c}	NGTG (77), NSTI (236), NDTN (307), NDTL (317)
	GRMZM2G015804		8.83	None ^b	
			8.83		
			8.90		

Table 1 continued

Group	Gene Name	Molecular weight in Daltons(number of residues)	Theoretical pI ProtParam EMBL SMS	Prediction of location	Predicted N-glycosylation sites (position amino acid)
4	ZmBGLU26	57110.3 Da (508 aa)	6.13 6.12 6.58	Secretory pathway ^{a,b,c}	NGTG (118), NSTD (348)
	GRMZM5G0828987				
	ZmBGLU9	33127.4 Da (290 aa)	6.20 6.20 6.68	Secretory pathway	NRSD (22), NITE (267)
	GRMZM2G021379				
	GRMZM5G845736				
	ZmBGLU22	57616.2 Da (511 aa)	6.09 6.07 6.51	Mitochondrial ^a Secretory pathway ^{b,c}	NKSL (61)
	GRMZM2G031660				
	ZmBGLU23	57747.8 Da (519 aa)	5.77 5.71 6.03	Secretory pathway ^{a,b} Vacuole ^c	NSTD (277)
	GRMZM2G457040				
	GRMZM5G810727				
	ZmBGLU24	64720.7 Da (562 aa)	6.04 6.02 6.46	Secretory pathway ^{a,b} Chloroplast ^c	NSST (217), NITE (266), NIST (408)
	GRMZM2G012236				
	ZmBGLU25	56268.3 Da (490 aa)	8.24 8.09 8.23	Chloroplast ^{a,c} Secretory pathway ^b	NRSD (22), NITE (271)
	GRMZM5G882852				

GRMZM gene names according to MaizeSequence database and suggested ZmBGLU nomenclature (see also Fig. 3). The theoretical isoelectric point (pI) was calculated with ProtParam (Gasteiger et al. 2005), with the EMBL pI tool which is based on (Lehninger 1979) and with Sequence Manipulation Suite (Stothard 2000). Full length protein sequences were analyzed for the presence of signal peptides for subcellular localization using ^a TargetP v1.1, ^b Predotar v1.03 and ^c WoLF Psort Prediction. Different bioinformatic algorithms can make distinct predictions since some proteins can have multiple targeting to different compartments. Putative N-Glycosylation sites were determined with the CBS prediction servers (<http://www.cbs.dtu.dk/services/>)

Table 1 have the TFNEP, ENG and EPY motifs, and therefore we assume that they could be catalytically active enzymes with β -glucosidase activity.

Analysis of the gene structure of the *ZmBGLU* family

Having separated the BGLU proteins into subgroups, a more extensive bioinformatic analysis of the maize *bglu* family was subsequently performed. The two previously annotated genes (*bglu1* and *bglu2*) which formed the core of group 1 had a very similar gene structure of 12 exons and 11 introns (Fig. 1). We then compared the genomic sequences of all *bglu* genes with the dot plot technique and found that sequence conservation of the introns is much lower for the other members of the family (data not shown). Nevertheless, the intron–exon structure displayed some regular patterns specific to protein groups (Fig. 4). The gene organization group 1 genes comprise 12 exons and 11 introns (Fig. 4). This exon structure is also typical of *bglu* genes in other species (Xu et al. 2004; Opassiri et al. 2006). The maize pattern is of one large exon, followed by 5 small, 2 large, 1 tiny and 3 medium size exons (Fig. 4). There were exceptions of *bglu* genes with less or more exons. The genes *bglu10* and *bglu19* had only 9 exons, while *bglu12* and *bglu20* had 13 exons (Fig. 4).

Characterization of the *Zmbglu* proteins

We also made some group specific protein alignments in order to identify group specific conserved motifs (data not

shown). The enzymes were also characterized with respect to the molecular weight, isoelectric point (pI), subcellular targeting (presence or absence of signal peptides) and N-glycosylation sites (Table 1). Using the orthologue tool of MaizeSequence database we determined for each *ZmBGLU* isoform the probable orthologs in other plant species (Table 2). We also blasted all the entries to the arabidopsis genome and found that each group of *ZmBGLU*s was most similar to different member of the arabidopsis *AtBGLU*s (data not shown).

With the exceptions of BGLU7 and BGLU10, all group 1 BGLU proteins are predicted to contain a plastid targeting sequence (Table 1). This suggests that group 1 enzymes can come into contact with substrates that are inside the plastid (e.g. in vivo release of active hormones), or after wounding by a pathogen by cell rupture and mixture of all subcellular compartments (Monroe et al. 1999). Most BGLUs of group 1 (except BGLU5 and BGLU7) contain at least one predicted N-Glycosylation site (Table 1). Glycosylation can be a way to inactivate enzymes for storage (Jones and Vogt 2001) or to make them more soluble for stability.

Compared to group 1, the other groups of *bglu* genes exhibit more variation of molecular properties. Not only do the *bglu* genes from other groups have a different gene structure (Fig. 4), but also the predicted cellular localization is rather directed to the secretory pathway and not limited to the plastid as in group 1 (Table 1).

Interestingly, the theoretical isoelectric point of some BGLUs correspond to the biological pH of the predicted

Fig. 3 Cluster dendrogram of β -glucosidase proteins in the B73 maize genome. Grouping was based on sequence identity at the protein level (Fig. 2). The numbering of group 1 was defined by the previously cloned genes *bglu1* and *bglu2* (yellow arrows). The other groups were defined according to a similarity threshold. BGLU proteins were numbered according to our iterative sequential analysis (Table 1). A beta-glucosidase gene family nomenclature is suggested in blue (*bglu1*–*bglu26*). Two proteins (BGLU9 and BGLU10) had a relatively low sequence similarity (~50%) to other members of their groups, but were nevertheless assigned to them

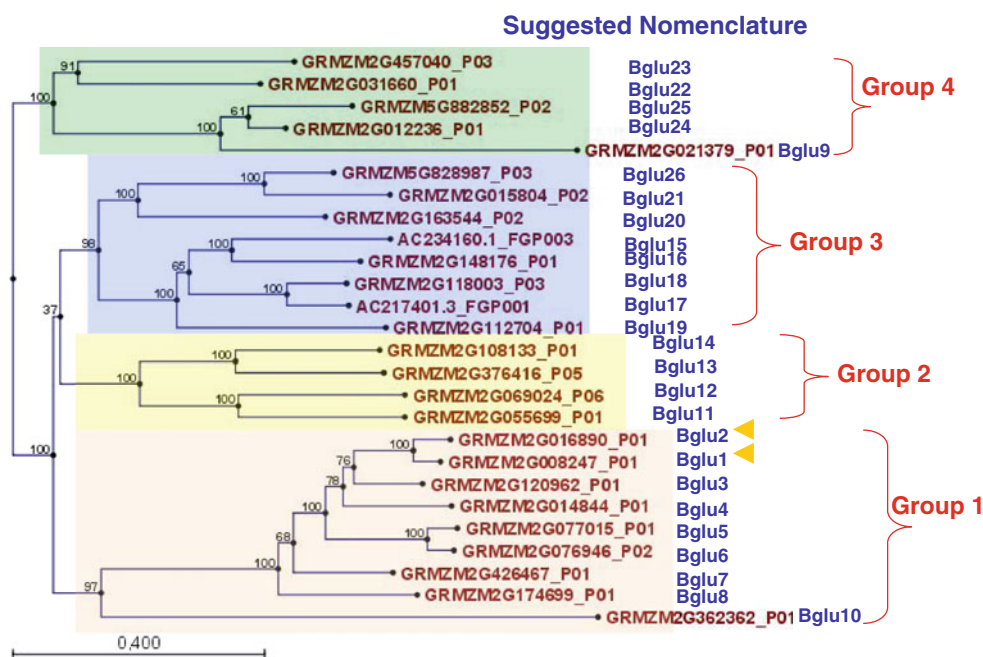
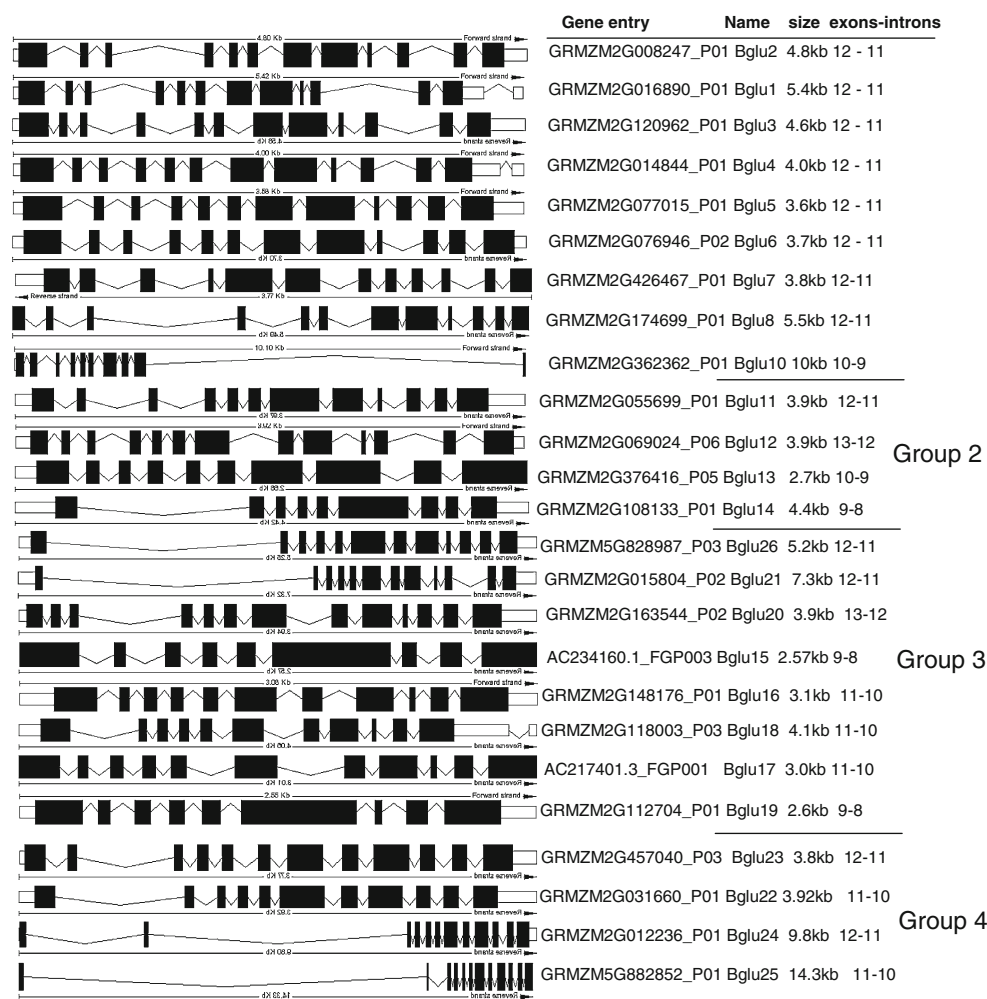


Fig. 4 Intron-Exon structure of β -glucosidase genes. Intron-exon structure of *bglu* genes. The name and size of the gene and number of exons and introns are indicated for each isoform



location. The average of the theoretical isoelectric point of group 1 proteins also compares to the chloroplast pH = 5 in thylakoid space, and pH = 8 in stroma (Orij et al. 2009). The isoelectric points of *BGLU20*, *BGLU23* and *BGLU26* are rather acidic, whereas *BGLU5*, *BGLU6*, *BGLU15* and *BGLU21* are rather alkaline (Table 1). This somehow corresponds to chemical milieu of the secretory pathway or plastid/mitochondria, respectively.

The group 2 BGLUs is characterized by acidic isoelectric points whereas group 3 is rather characterized by alkaline isoelectric points (Table 1). Group 4 has rather neutral isoelectric points (Table 1).

The group 2 and 3 proteins are characterized by having three to five N-glycosylation sites, whereas groups 1 and 4 have much less predicted N-glycosylation sites.

Some genes display dual targeting, for example, *BGLU11* and *BGLU13* were predicted to have dual targeting to the mitochondria/vacuole, *BGLU12* and *BGLU14* have dual targeting to the plastid/vacuole, whereas *BGLU5* and *BGLU15* are both targeted to mitochondria or plastids (Fig. 8).

Analysis of the gene promoter regions

In addition to the protein coding regions, the promoter regions of all *bglu* genes were also analyzed for the presence of *cis* elements. This was done by selecting the genomic sequences ~550 bp upstream of the protein start site. Many consensus sequences for DNA binding proteins were found, some of them related to transcription factors mediating responses to biotic and abiotic stress. The TATA box element for transcriptional machinery assembly was found in 18 of the *bglu*'s. Some genes (namely *bglu* 7, 14, 17, 18, 20 and 22) did not contain any of the known TATA box elements (TATA Box1, 2, 3, 4, 5, TATA Box OSPAL, TATAPVTRNALEU) which are believed to be critical for accurate initiation (Grace et al. 2004). However, all of the *bglu*'s promoter regions contain binding sites of at least one *cis* element involved in transcriptional activation of several defense-related genes, regulated by hormones or elicitors (Table S1). For example, we found auxin elements, salt-induced, heat shock, jasmonate signalling and antioxidant response (Table S1). Twenty *bglu* genes have *cis* elements

Table 2 List of BGLU orthologs in other plant species

Group Maize	Protein name <i>Zea mays</i>	<i>Sorghum bicolor</i>	<i>Bachipodium dystachyon</i>	<i>Oryza sativa</i>	<i>Arabidopsis thaliana</i>
1	ZmBGLU1	Sb08g007570	None displayed in release 5b.60 (ND)	None displayed in release 5b.60 (ND)	AT1G51470 AtBGLU35
	GRMZM2G016890_P01	Sb08g007586			AT1G51490 AtBGLU36
	Chr 10	Sb08g007610			AT5G25980 AtBGLU37
		Sb08g007650			AT5G26000 AtBGLU38
					AT5G48375 AtBGLU39
					AT1G47600 AtBGLU34
	ZmBGLU2	Sb08g007570	ND	ND	AT1G51470 AtBGLU35
	GRMZM2G008247_P01	Sb08g007586			AT1G51490 AtBGLU36
	Chr 10	Sb08g007610			AT5G25980 AtBGLU37
		Sb08g007650			AT5G26000 AtBGLU38
					AT5G48375 AtBGLU39
					AT1G47600 AtBGLU34
	ZmBGLU3	Sb08g007570	ND	ND	AT1G51470 AtBGLU35
	GRMZM2G120962_P01	Sb08g007586			AT1G51490 AtBGLU36
	Chr 3	Sb08g007610			AT5G25980 AtBGLU37
		Sb08g007650			AT5G26000 AtBGLU38
					AT5G48375 AtBGLU39
					AT1G47600 AtBGLU34
	ZmBGLU4	Sb08g007570	ND	ND	AT1G51470 AtBGLU35
	GRMZM2G014844_P01	Sb08g007586			AT1G51490 AtBGLU36
	Chr 10	Sb08g007610			AT5G25980 AtBGLU37
		Sb08g007650			AT5G26000 AtBGLU38
					AT5G48375 AtBGLU39
					AT1G47600 AtBGLU34
	ZmBGLU5	Sb08g007570	ND	ND	AT1G51470 AtBGLU35
	GRMZM2G077015_P01	Sb08g007586			AT1G51490 AtBGLU36
	Chr 3	Sb08g007610			AT5G25980 AtBGLU37
		Sb08g007650			AT5G26000 AtBGLU38
					AT5G48375 AtBGLU39
					AT1G47600 AtBGLU34
	ZmBGLU6	Sb08g007570	ND	ND	AT1G51470 AtBGLU35
	GRMZM2G076946_P02	Sb08g007586			AT1G51490 AtBGLU36
	Chr 3	Sb08g007610			AT5G25980 AtBGLU37
		Sb08g007650			AT5G26000 AtBGLU38
					AT5G48375 AtBGLU39
					AT1G47600 AtBGLU34
	ZmBGLU7	Sb08g007570	ND	ND	AT1G51470 AtBGLU35
	GRMZM2G426467_P01	Sb08g007586			AT1G51490 AtBGLU36
	Chr 3	Sb08g007610			AT5G25980 AtBGLU37
		Sb08g007650			AT5G26000 AtBGLU38
					AT5G48375 AtBGLU39
					AT1G47600 AtBGLU34
	ZmBGLU8	Sb08g007570	ND	ND	AT1G51470 AtBGLU35
	GRMZM2G174699_P01	Sb08g007586			AT1G51490 AtBGLU36
	Chr 10	Sb08g007610			AT5G25980 AtBGLU37
		Sb08g007650			AT5G26000 AtBGLU38
					AT5G48375 AtBGLU39
					AT1G47600 AtBGLU34
	ZmBGLU10	Sb06g019880	BRADI5G13260	LOC_Os04g39880 LOC_Os04g39900	AT2G44450 AtBGLU15
	GRMZM2G362362_P01				AT3G60130 AtBGLU16
	Chr 2				AT5G42260 AtBGLU12
					AT5G44640 AtBGLU13
					AT2G25630 AtBGLU14

Table 2 continued

Group Maize	Protein name <i>Zea mays</i>	<i>Sorghum bicolor</i>	<i>Bachipodium dystachyon</i>	<i>Oryza sativa</i>	<i>Arabidopsis thaliana</i>
2	ZmBGLU11	Sb09g018180	BRADI2G27770	LOC_Os05g30350	AT1G45191 AtBGLU1
	GRMZM2G055699_P01	Sb06g022440	BRADI2G09187	LOC_Os05g30390	AT1G60090 AtBGLU4
	Chr 6		BRADI2G09200		AT1G60260 AtBGLU5
					AT1G60270 AtBGLU6
					AT3G62740 AtBGLU7
					AT3G62750 AtBGLU8
					AT4G22100 AtBGLU3
					AT4G27820 AtBGLU9
					AT4G27830 AtBGLU10
					AT5G16580 AtBGLU2
					AT1G02850 AtBGLU11
	ZmBGLU12	Sb03g037780	BRADI2G59650	LOC_Os01g70520	AT1G45191 AtBGLU1
	GRMZM2G069024_P06	Sb06g022440	BRADI2G59660		AT1G60090 AtBGLU4
	Chr 3		BRADI2G09187		AT1G60260 AtBGLU5
			BRADI2G09200		AT1G60270 AtBGLU6
			BRADI2G09260		AT3G62740 AtBGLU7
			BRADI3G45607		AT3G62750 AtBGLU8
			BRADI3G45627		AT4G22100 AtBGLU3
			BRADI3G45640		AT4G27820 AtBGLU9
			BRADI3G45650		AT4G27830 AtBGLU10
					AT5G16580 AtBGLU2
					AT1G02850 AtBGLU11
	ZmBGLU13	Sb10g022300	BRADI4G34940	LOC_Os09g33690	ND
	GRMZM2G376416_P05	Sb06g022440	BRADI4G34950	LOC_Os09g33710	
	Chr 9				
	ZmBGLU14	Sb02g029640	BRADI4G34927	LOC_Os09g33680	ND
	GRMZM2G108133_P01	Sb06g022440			
	Chr 7				
3	ZmBGLU15	ND	ND	ND	ND
	AC234160.1_FGP001				
	Chr 8				
	ZmBGLU16	Sb02g041550	BRADI1G19270	LOC_Os07g46280	AT3G18080 AtBGLU44
	GRMZM2G148176_P01				AT3G18070 AtBGLU43
	Chr 7				
	ZmBGLU17	Sb01g010830	BRADI1G10890	LOC_Os03g49600	AT3G18080 AtBGLU44
	AC217401.3_FGP001		BRADI1G10917		AT3G18070 AtBGLU43
	Chr 1		BRADI1G10930		
	ZmBGLU18	Sb01g010840	BRADI1G10890	LOC_Os03g49600	AT3G18080 AtBGLU44
	GRMZM2G118003_P03		BRADI1G10917		AT3G18070 AtBGLU43
	Chr 1		BRADI1G10930		
	ZmBGLU19	Sb10g028060	BRADI4G08040	LOC_Os12g23170	AT3G18080 AtBGLU44
	GRMZM2G112704_P01				AT3G18070 AtBGLU43
	Chr 5				
	ZmBGLU20	Sb10g027600	BRADI1G33040	LOC_Os06g46940	AT5G54570 AtBGLU45
	GRMZM2G163544_P02				
	Chr 5				
	ZmBGLU21	Sb01g043030	BRADI1G70170	LOC_Os03g11420	AT1G26560 AtBGLU40
	GRMZM2G015804_P02				
	Chr 1				
	ZmBGLU26	ND	ND	ND	ND
	GRMZM5G0828987_P03				
	Chr 9				

Table 2 continued

Group Maize	Protein name <i>Zea mays</i>	<i>Sorghum bicolor</i>	<i>Bachipodium dystachyon</i>	<i>Oryza sativa</i>	<i>Arabidopsis thaliana</i>
4	ZmBGLU9	ND	ND	ND	ND
	GRMZM2G021379_P01				
	Chr 2				
	ZmBGLU22	Sb06g022500	ND	LOC_Os04g43380	AT1G61820 AtBGLU46
	GRMZM2G031660_P01	Sb06g022510		LOC_Os04g43410	AT4G21760 AtBGLU47
	Chr 10				AT1G61810 AtBGLU45
	ZmBGLU23	ND	ND	ND	ND
	GRMZM2G457040_P03				
	Chr 10				
	ZmBGLU24	Sb06g022385	BRADI5G15527	LOC_Os04g43360	AT1G61820 AtBGLU46
	GRMZM2G012236_P01	Sb06g022390			AT4G21760 AtBGLU47
	Chr 2	Sb06g022400			AT1G61810 AtBGLU45
		Sb06g022410			
		Sb06g022420			
		Sb06g022450			
		Sb06g022460			
	ZmBGLU25	Sb06g022385	BRADI5G15527	LOC_Os04g43360	AT1G61820 AtBGLU46
	GRMZM5G882852_P02	Sb06g022390			AT4G21760 AtBGLU47
	Chr 2	Sb06g022400			AT1G61810 AtBGLU45
		Sb06g022410			
		Sb06g022420			
		Sb06g022450			
		Sb06g022460			

potentially regulated by ABA (except *bglu7*, 8, 20 and 21) and all *bglu*'s have *cis* elements which are related to dehydration or light responses (Table S1).

Some very specific *cis* elements linked to ethylene and ammonium regulation were found in *bglu4*, 5, 7, 11, 13 and 16 (Table S1). The genes *bglu6*, 8, 9, 11, 12, 13, 14, 15, 16, 18, 19, 21 and 23 have some *cis* elements that are regulated by low temperature (Table S1).

In general, *in-silico* analysis suggested that the promoter regions of the *Zmbglu*'s genes have a distinct set of *cis* elements and transcription factor binding sites. In order to visualize the occurrence of *cis* element in the *bglu* genes more clearly, the data of Table S1 was used to construct a heatmap bicluster (Fig. S5). The dendrogram on the side of the heatmap indicates the degree of identity between the *bglu* genes with respect to the regulatory elements in their promoter regions (Fig. S5). The dendrograms of Fig. 3 and Fig. S5 cluster the genes differently, meaning that there is little or no correspondence between sequence similarity at the coding region (protein), and sequence similarity at the promoter region.

We then focused on the promoters of *bglu1* and *bglu2*, looking for evidence that the *cis* elements are responsible for β -glucosidase activity induced by stress in maize. Specifically six specific sequences in *bglu2* (elicitors induced, like: BOXLCOREDPCAL, SEBFCONSSTPR10A, DPBFCOREDPC3; ABA induced, like: DPBFCOREDPC3,

ABREOSRAB21, DPBFCOREDPC3) were detected, which were absent in *bglu1*. Also we found specific *cis* elements in *bglu1* absent in *bglu2*, like GCCCORE, T/GBOXATPIN2, involved in regulating jasmonate and pathogen-responsive gene expression, WBOXNTCHN48 involved in elicitor-responsive transcription of defense genes, RYREPEATBNNAPA related with ABA response and MYCATRD22 related with dehydration and ABA-induction responsive gene (Table S1).

In summary, we concluded that the *bglu* genes contain regulatory sequences that would modulate transcriptional activity in response to a variety of factors. In order to verify this hypothesis, we performed microarray analysis of all *bglu* genes on a set of tissues under various environmental conditions using different genotypes grown in different locations.

Microarray analysis of the *bglu* gene family

In order to obtain the expression values of the *bglu* gene family members separately, the data we obtained from the maize array service (University of Arizona) had to be analyzed as indicated in the method section and here briefly summarized. We first normalized the intensity spot data of all arrays and treated each dye (color channel) independently. We also made averages across replicas and slides. We then applied a special algorithm based on least square

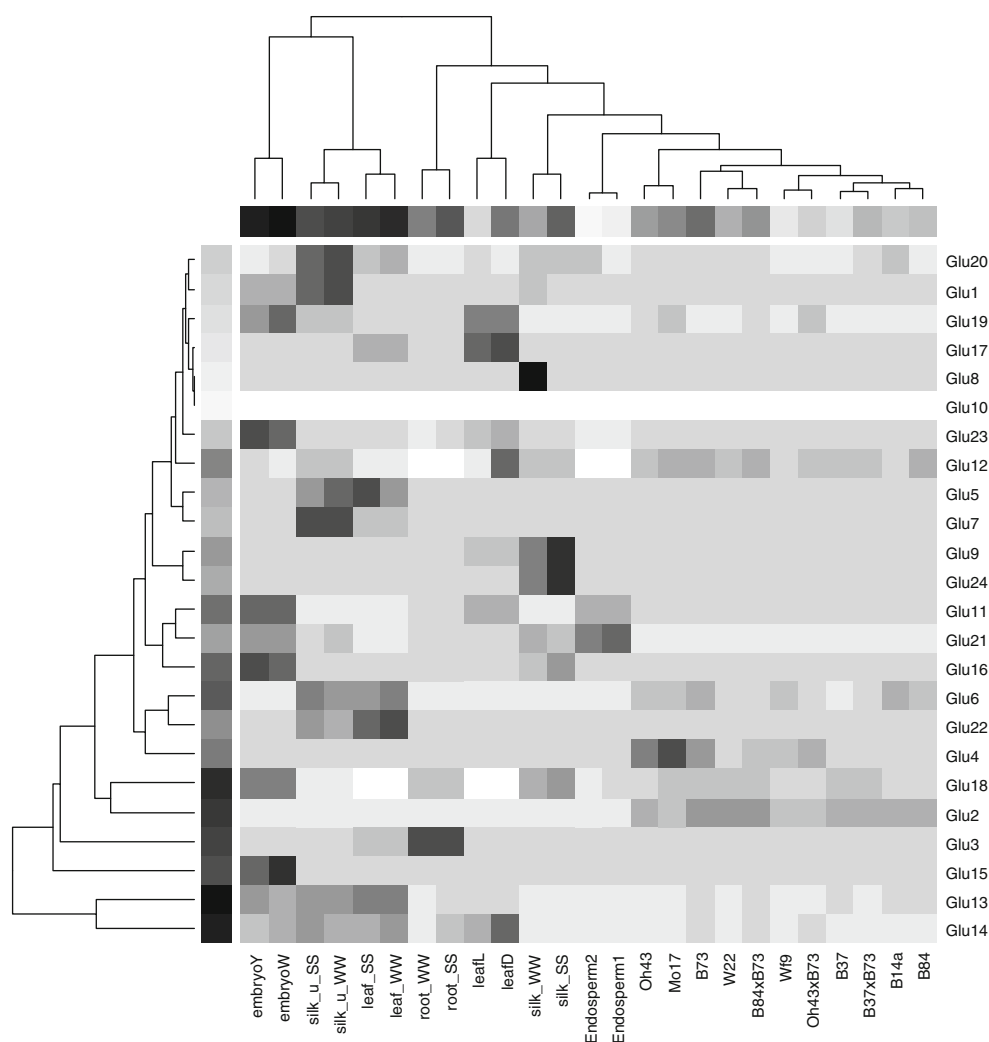


Fig. 5 Heatmap visualization of the expression pattern of all *bglu*'s genes in maize. Details of microarray experiments and statistical procedures are given in methods. The top and left dendrograms indicate clustering of samples or genes according to the expression profile. The legend indicates the corresponding names of the samples (at the bottom) or genes (on the right). RNA was extracted either from 15 day whole plantlets of different genotypes (B73, Mo17, W22, etc.) or different tissues from a pool of genotypes (silks, leaves, roots, endosperm, embryos, etc.) under different environmental conditions:

drought stress (SS), well watered (WW), Light (L), Dark (D). The left and top graybars (without label) show averaged levels of expression across rows or columns. The expression intensity is given in gray colors: Black means high expression (high intensity values of corresponding oligo probes in microarray), whereas white indicates low expression levels (low intensity values). For *bglu10* we could not identify any suitable oligo probe in the Arizona Maizearray and therefore is shown as blank

sums (see methods) in order to deconvolute the microarray data from values based on oligo probe spots (MZ entries) to values based on maize genes (GRMZM entries) (Fig. S2). The method is a best-fit approach that makes a mathematical simulation of theoretical expression values using the available experimental data as starting point. As any other statistical modeling method, it is highly susceptible to value variation and experimental errors, thus results interpretation has to be done with care. However, it nevertheless allowed the discrimination of the expression levels of individual *bglu* genes. The applied procedures (averages across slides, and averages across MZ probes) reduced the variance of the data. The averaged data of all microarray

experiments is summarized on Fig. 5. The expression profile of the *bglu* gene family is presented as a color-coded heatmap-biclustrogram (Fig. 5) that shows two clusters each for samples and for genes, respectively. This type of analysis allowed us identifying the genes that have a similar expression pattern across tissues and experimental conditions. We found no correspondence between the dendrogram of protein sequence similarity (Fig. 3) and the dendrogram of gene expression similarity (Fig. 5). This is indicative that similar BGLU proteins can have a very distinct expression pattern, while dissimilar BGLU proteins can be similarly expressed. This is in agreement with the view that none of the paralogs are redundant, but each one

might have a different function. The gene with most similar expression pattern to *bglu1* was *bglu20* (Fig. 5). Those genes have 35% equal *cis* elements in their sequence (Table S4). There are also other group of genes that share a similar transcriptional profile, namely the pair *bglu9* and *bglu20* and the pair *bglu5* and *bglu7* (Fig. 5), sharing 32 and 37% of *cis* elements in their sequence respectively (Table S1). However, there was only a low overall correspondence between the dendrogram of gene expression (Fig. 5), the dendrogram of protein identity (Fig. 3) and the dendrogram of *cis* regulatory elements (Fig. S5).

Taking into account the net intensity across all samples (see side-bars on Fig. 5), the genes with the highest values were *bglu18*, 13 and 14, whereas the lowest expression was measured for *bglu8*, 17 and 19. Interestingly, some *bglu* genes are higher expressed only under a particular set of conditions (Fig. 5). For example, the highest expression of *bglu3* was found on the roots, *bglu15* is expressed preferentially in the seed embryo, *bglu7* in the silks and *bglu5* both in silks and leaves. *bglu24* is expressed in silks and it is induced under severe stress (SS) in comparison to well watered (WW) samples. Expression of those genes in silks was only seen in greenhouse grown plants in Switzerland (silk samples), and not in field grown plants in Tlaltizapan (silk_u samples). The promoter region analysis of *bglu9* and *bglu24* found a GATA Box involved in light regulated and tissue specific expression and a ACGTATERD1 *cis* element required for etiolation-induced expression of *erd1* (early responsive to dehydration) and *MYCCONSENSUSAT* *cis* element involved in recognition site found in the promoters of the dehydration-responsive gene *rd22* and many other genes in arabidopsis. *Zmbglu24* have a CuRE (copper-response element), also involved in oxygen-response (CURECORECR) and an activator of the carrot phenylalanine ammonia-lyase gene (*DcPAL1*) in response to elicitor treatment, UV-B irradiation (BOXLCOREDPCAL).

A similar observation was done for other genes. For example, *bglu17* was mainly expressed in mature leaf tissue, but expression was higher in plants grown in the greenhouse in Irapuato (leafL and leafD samples), much less in plants grown in Tlaltizapan (leaf_SS and leaf_WW samples).

The previously characterized gene *bglu1* was only weakly expressed in almost all conditions and tissues (mainly mature plants), but it was higher in silk samples of field grown plants, and it was slightly repressed by water stress. In contrast, *bglu2* was on average of all tissues much higher expressed than *bglu1* (Fig. 5). It was low in silks and embryos and almost all other mature tissues, whereas expression was higher in 15 day plantlets of different genetic backgrounds (typical maize inbreds and hybrids from USA). There was also variation of *bglu* gene expression among the temperate genotypes. For example,

bglu4 was most highly expressed in the inbreds Mo17 and Oh43 whereas expression was lowest in W22, B37, B14a and B84 (Fig. 5).

When analyzing the different samples shown in Fig. 5, it becomes clear that samples from the same tissue cluster together, independently of the environmental treatment (e.g. water stress, light, yellow or white embryo, or opaque or normal endosperm). This is visualized by the cluster pairs of samples that correspond to the same tissue exposed to very different conditions (stress or light). This means that gene expression variance of the *Zmbglu* gene family is more influenced by the type of tissue and less by the environment or treatment.

The cluster analysis also allows comparing the environmental and the genetic variances. The genotype Oh43 clusters together with Mo17, whereas B73 clusters with W22 (Fig. 5). Incidentally, this corresponds also to the heterotic groups to which these genotypes belong. Since all temperate genotypes (lines and hybrids) cluster together and separately from the other samples (Fig. 5), it can be concluded that variance due to different genetic backgrounds (even from different heterotic groups) is relatively small in comparison to the variance due to the environment. The temperate genotype B73 grown in the USA clusters together with other temperate genotypes because they were all grown under similar conditions in USA and corresponded to the same tissues (15d old seedlings). The gene expression pattern was very different to the B73 samples grown in México because fully expanded green leaves of mature plants were harvested (leafL and leafD samples in Fig. 5) and not whole plantlets as in the USA samples.

Spatial-temporal gene expression of *bglu1* and *bglu2* during seed germination

β -glucosidase is the most abundant protein in soluble coleoptile extracts (Esen 1992), but the question emerged to whether the activity is encoded only by *bglu1* or by *bglu2*. Previous reports suggested that *bglu1* is mainly expressed in coleoptiles of maize, whereas the *bglu2* gene has been reported to be expressed exclusively in maize leaves at low levels and only 6 days after germination (Esen and Blanchard 2000). The expression of *bglu1* and *bglu2* in coleoptiles of 3, 4, 5 and 6-days-old germinated seeds and in leaves and stem of 7 and 10-days-old germinated seeds, was analyzed by RT-PCR using gene-specific primers under optimized conditions for the two β -glucosidase genes (Fig. S3). *Bglu1* and *bglu2* mRNA was detected in all samples (Fig. 6). Interestingly, *bglu1* and *bglu2* expression level decayed (around 60%) in stems from 7-days post-germinated seeds. *Bglu1* mRNA levels were relatively low and constant throughout the experiment

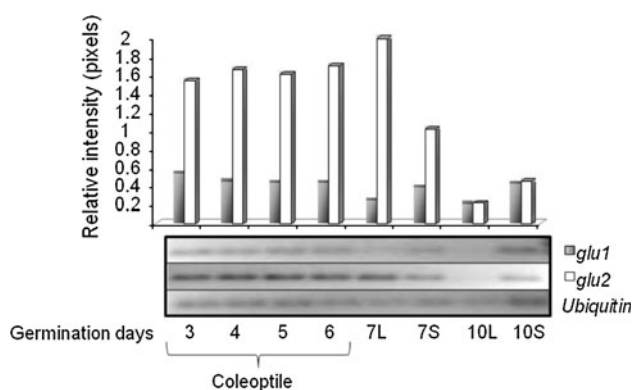


Fig. 6 Spatio-temporal of *bglu1* and *bglu2* gene expression in the B73 genotype. Relative intensity of *bglu1* (graybox) and *bglu2* (white box), the samples of 3, 4, 5 and 6-day old germinated seeds are coleoptiles and 7 and 10-day old germinate seeds are leaves (L) and stem (S). Germination was done on wet filter paper at 28°C in the light

and *bglu2* transcript levels were detected from day 3 to day 7 post-germination with maximum expression in coleoptiles (Fig. 6).

bglu1 and *bglu2* expression under salt stress or mechanical damage

In salt stressed maize plants *bglu1* mRNA levels changed with a particular transient pattern, in the first 30 min the mRNA levels increase with a maximum significant increase (ANOVA, $P = 0.03$) at 1 h. After 3 h the

expression decreased to 60% of normal levels, and after 24 h the mRNA level had returned to the starting value (Fig. 7). This pattern of rapid induction after stimulus, with a transient expression peak and then recovering to normal levels is typical of early stress responsive genes. For the *bglu2* a tendency to decrease was observed under both treatments. Changes in *bglu1* mRNA levels in response to mechanical damage were not significant. The *bglu1* mRNA levels were practically constant during 24 h whereas *bglu2* had a tendency to decrease to 51% already after 1 h of stress compared with the initial values (Fig. 7).

β -glucosidase activity in different maize tissues

In addition to RNA levels, we also measured the total BGLU enzymatic activity. In order to investigate the relationship between *bglu1* and *bglu2* mRNA levels and enzymatic activity following salt stress in maize, we evaluated BGLU activity in coleoptile homogenates (Maize, 7 post-germination days) under saline stress (300 mM NaCl) and observed an increase of 3 fold in 30 min after salt stress (Fig. S4).

We also measured the BGLU activity on the different tissues of seedlings after several germination days (Table 3). It seems that the total activity in the plant increases with germination during the first week. Maximum activity is in the coleoptile/stem before leaf expansion (day 4) and then switches to the leaves with some remaining activity on the stems. Root and seeds have lower

Fig. 7 Relative intensity of *bglu1* and *bglu2* gene expression in salt stress and mechanical damage. The data of relative intensity (pixels) are shown with the mean and the standard error

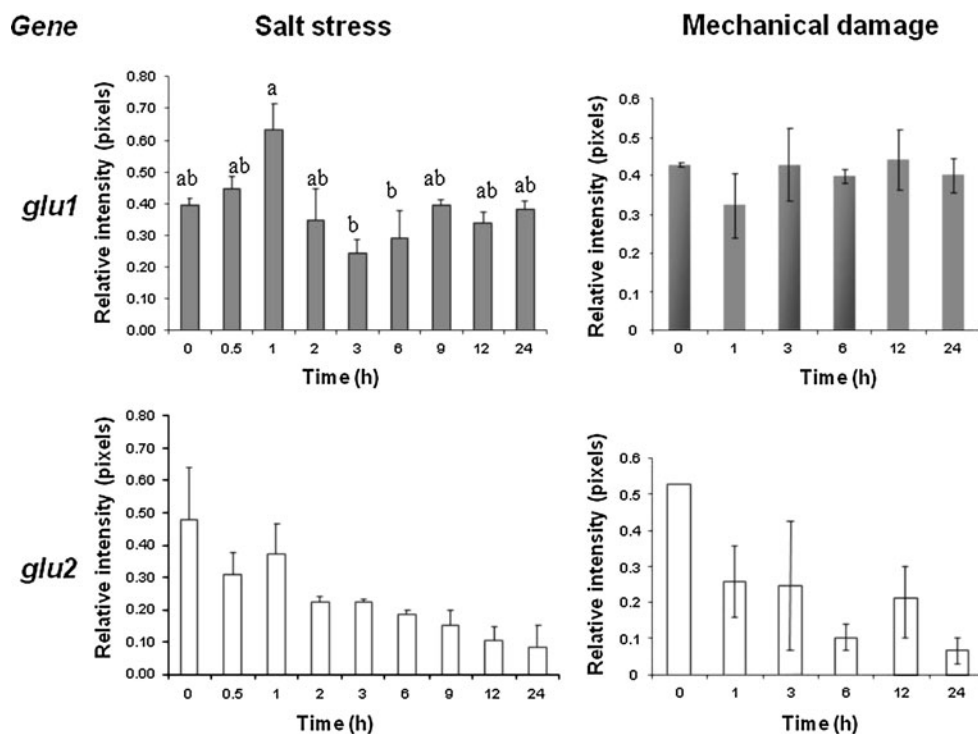


Table 3 β -Glucosidase activity in different tissues after germination

	0d	3d	4d	6d	7d	10d
Seed	0.193	0.441	0.072	0.796	0.132	0.283
Coleoptile	ND	0.269	1.712	0.050	0.396	0.122
Leaf	ND	ND	ND	0.813	1.048	0.508
Root	ND	0	0.755	0.667	0.434	0.421
Total	0.193	0.710	2.539	2.326	2.010	1.334

Table 4 β -Glucosidase activity in different tissues of maize

Tissue	BGLU activity
Seed (mature dry seeds)	0.01 $\pm$ 0.01
Coleoptiles (5-day old seedlings)	1.423 $\pm$ 0.011
Leaf blade (65-day old plants)	0.715 $\pm$ 0.003
Leaf sheath (65-day old plants)	0.288 $\pm$ 0.008
Husk leaves (65-day old plants)	0.260 $\pm$ 0.030
Ear (2-day after pollination)	0.141 $\pm$ 0.006
Silks (2-day after pollination)	1.344 $\pm$ 0.078

β -Glucosidase activity was measured in different tissues of the maize inbred B73. The values indicate means $\pm$ SE in arbitrary units of activity (OD 410 nm) per gram dry weight

activity levels than in coleoptile and leaf extracts. The activity results resembles those obtained from the mRNA expression levels and suggest that the half life of the active enzyme is not long, since the activity decreases shortly after the expression decreases too.

We also measured the BGLU activity in the different tissues of mature plants (Table 4). BGLU activity was low in tissues like maize ears and husk leaves. Leaf blades contain higher BGLU activity whereas silks presented the highest activity of all tissues analysed (Table 4). The genes that are preferentially expressed in leaves are *bglu14*, *bglu11*, *bglu12*, *bglu17* and *bglu19* whereas the genes preferentially expressed silks are *bglu18*, *bglu16*, *bglu24*, *bglu9*, and *bglu8* (Fig. 5).

Discussion

Databases on maize genetics and genomics are not yet coordinated and sometimes use a non-standard gene nomenclature

The public databases such as MaizeGDB contain large amounts of useful information. Most of the data is correct, but it also contains erroneous information that can mislead various users, including molecular plant breeders searching

for accurate map locations. Sometimes ambiguous molecular markers are used for gene mapping, particularly when dealing with large gene families. Other times, non-standard nomenclatures are used for some genes and enzymes. Therefore, it is important to examine the data across different databases for each individual gene family. We did a manual curation of the published information for the maize genes *glu1* and *glu2* (here renamed to *Zmbglu1* and *Zmbglu2*). We found that *glu2* is not located on chromosome 2 as currently reported in the MaizeGDB webpage. In the MaizeSequence database it can be consulted that *bglu1* and *bglu2* are both located on chromosome 10 (Table 2). However, the MaizeSequence database currently lists only two genes that are annotated as β -glucosidase enzymes (BGLU).

Identification of all paralogs for a given enzyme family requires manual curation of the genomic data

The automatic procedures that are currently used for annotating gene paralogs in the MaizeSequence database is not sufficient to identify all member of a given gene family. For example, in the recent release 5b.60, the gene GRMZM2G016890 (*bglu1*) lists only 16 paralogs. This putative list of paralogs contains mainly BGLUs from group 1 and 3, however it does not include members of group 2 and group 4 BGLUs that can be selected manually via protein Blast.

It is also important to distinguish full length enzymes from pseudoproteins or truncated proteins. For example, the protein GRMZM2G119460_P01 has a high local similarity to BGLU1 and it contains a Glyco_hydro_catalytic_core, but it codes only for a protein of only 67 aa and therefore is far from being a functional TIM-Barrel enzyme with a typical sequence length ($\sim$ 450 aa). Another example is the gene GRMZM2G031693 that it codes only for a protein of 256 aa, which is almost half in size as all other BGLU isozymes. Furthermore, it is important to check that the paralogues contain the typical BGLU peptides such as TFNEP, EPY and ENG that are characteristic for all functional BGLU enzymes (Fig. 2). For example, since BGLU9 and BGLU10 do not actually contain the

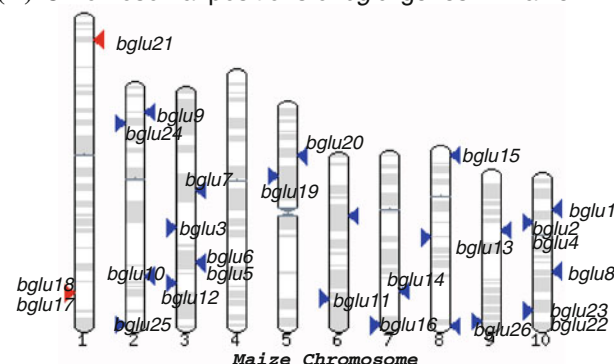
ENG motif, we postulate that they are either non-functional enzymes, or that they have a very different catalytic mechanism as all the other BGLUs. Molecular characterization of enzyme families in maize requires extensive bioinformatic analysis including iterative steps and manual curation of the nucleotide and protein data as described in this article.

Analysis of enzyme families in maize must be revised in future genomic releases

In plants genomes, it is typical that some key metabolic enzymes are represented by several proteins or isozymes that have slightly different kinetic properties (substrate specificity and regulatory properties). In arabidopsis, 47 *bglu* genes have been annotated (*Atbglu*) (Xu et al. 2004). In rice, 34 *bglu* genes are apparently functional (*Osbglu*) (Opassiri et al. 2006). A targeted gene survey was started in order to map and characterize all *bglu* paralogs in maize (*Zmbglu*). We found most members of β -glucosidase family in the maize genome by doing several BLAST queries using the PFAM domain (Glucosyl hydrolase) and the protein sequences of BGLU1 (=GRMZM2G016890_P01) and BGLU2 (=GRMZM2G008247_P01) (Fig. S1). There are copies of BGLU sequences on all maize chromosomes besides chromosome 4 (Fig. S1 and Fig. 8). After manual curation of all genomic entries of the release 4a.53 and the release 5b.60, we obtained a final list of 26 unique BGLU proteins (Table 1), from which 24 contain the key amino acid residues for catalytic function (Fig. 2) and that therefore, we believe to be putatively functional as enzymes (see also below).

When comparing different genomic releases, two BGLU proteins were additionally identified in release 5b.60, but we also found four genes that have been excluded, namely GRMZM2G021379 (BGLU9), AC234160.1_FG003 (BGLU15), AC217401.3_FG001 (BGLU17) and GRMZM2G457040 (BGLU23). There could be various reasons for this. For example, the protein GRMZM2G457040_P03 (BGLU23) of release 4a.53 is actually identical with the newly annotated protein GRMZM5G810727_P02 in release 5b.60. The protein GRMZM2G021379_P01 (BGLU9) of release 4a.53 corresponds to the newly annotated protein GRMZM5G845736_P01 in release 5b.60. Thus it seems that some maize enzymes such as BGLU9 and BGLU23 might simply change the GRMZM entry in the new genomic releases. We think that the proteins that we originally numbered as BGLU15, BGLU17 in the 4a.53 release (Table 1) were mistakenly excluded as annotated proteins in the filtered gene set of the release 5b.60. It can be expected that a few other BGLU genes will be added or renamed as the MaizeSequence database is updated. Nevertheless, it is already evident that cereal crops such as maize, sorghum and

(A) Chromosomal positions of *bglu* genes in maize



(B) Subcellular targeting of BGLU proteins

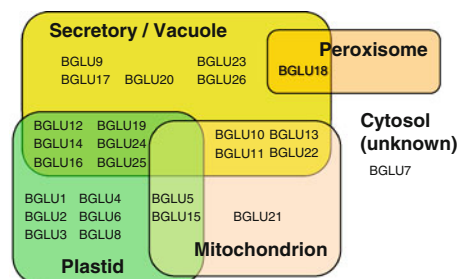


Fig. 8 Summary of the *Zea mays bglu* gene family. **a** Genetic positions of *bglu* genes in the different maize chromosomes. See Table 1 and Fig. 2 for the assignment of genes and the exact physical positions. **b** Subcellular targeting of BGLU proteins. Putative dual targeting of proteins is indicated for the isoforms that are drawn at the intersection of rectangles (see also Table 1)

rice (Opassiri et al. 2006) contain less BGLU paralogs than arabidopsis (Xu et al. 2004).

There are no two *Zmbglu* paralogs that have the same molecular properties, thus the rate of gene subfunctionalization in maize is more rapid than the rate of gene duplication

Among all BGLU's, the most similar pairs of proteins were BGLU1-BGLU2 (~91% identity), BGLU5-BGLU6 (~92% identity) and BGLU9-BGLU25 (~92% identity) and therefore cluster together (Fig. 3). The similarity between *bglu1* and *bglu2* is also evident at the intron and promoter regions (Fig. 1), suggesting a relatively recent duplication of those genes. Usually, enzyme families are enlarged by the occurrence of multiple isoforms that acquire different functions or are expressed differentially. Therefore, in addition to characterizing the gene structure (Fig. 4), we also studied the putative subcellular targeting and isoelectric point (pI) values of all maize BGLU proteins (Table 1). We conclude that there are no two *bglu* isogenes that have the same molecular properties. We also found that pI values and subcellular targeting correlate in some cases.

The high sequence divergence across the maize BGLU proteins suggests that most *Zmbglu* genes seem to have emerged much earlier than the domestication of maize or the evolution of cereals. The maize genome has undergone a relatively recent endoduplication (Schnable et al. 2009), however, for the maize *bglu* gene family there are no such marked pairs of similar proteins (Fig. 3). Even for the pair *bglu5-bglu6* they are still distinguishable at the genomic level (data not shown). The pairs of paralogs *bglu1-bglu2*, *bglu5-bglu6* and *bglu17-bglu18* are not even in syntenic positions of the endoreplicated maize chromosomes (Schnable et al. 2009) but rather correspond to nearby tandem gene duplications in the same DNA strand (Fig. 8). The only pair of similar paralogs that we found on syntenic positions was *bglu21* at the top of chromosome 1 and *bglu26* at the bottom of chromosome 9 (Fig. 8). Thus, we believe that the rate of BGLU protein subfunctionalization must be more rapid than the rate of gene duplication, a feature that is typical of genes/enzymes that are under strong selective pressure favoring the diversification of the sequences/functions.

There are some domains that are conserved in all BGLU enzymes and some residues that are characteristic for some isoenzymes

The glutamate residues in the motifs **TFNEP** and **IXENG** are conserved in all BGLU proteins and correspond to key amino acids involved in catalytic breakage of glucosidic bonds (Czjzek et al. 2000; Zouhar et al. 2001). This is in support of the view that most of the selected *ZmBGLU* proteins (Table 1) are functional enzymes with β -glucosidase activity (with the exception of BGLU9 and BGLU10). Despite having a similar catalytic function (hydrolysis of beta-glycosidic bonds), the *ZmBGLU* enzymes might differ in their substrate specificity or binding properties. For example, it has been reported that both lysine and threonine within the protein motif **YHMYKTDV** (see alignment position 140 in Fig. 3; see also Fig. S6) are involved in binding to the protein beta-glucosidase aggregating factor (BGAF) (Yu et al. 2009b). The key amino acid pair KT is only conserved in BGLU1 (GRMZM2G016890) and BGLU2 (GRMZM2G008247) but not in all other BGLU isoforms that rather contain **YHRYKEDV** as consensus (Fig. 2). Thus, it seems that both BGLU1 and BGLU2 might have the ability to bind to BGAF but not the other BGLU isoforms.

Structural data from co-crystals of enzyme-substrate and enzyme-aglycone complexes (see pdb entry 1e4n) have shown that five amino acid residues (F198, F205, W378, F466, and A467) are located in the aglycone-binding site of BGLU1 and form the basis of aglycone recognition and binding, hence substrate specificity. Furthermore, it has

been reported that the enzyme specificity towards aryl ring glucoside substrates is determined by the aglycone aromatic system stacking with W373, and interactions with edges of F193, F200, and F461 (see pdb entry 1hxj) (Zouhar et al. 2001; Verdoucq et al. 2003). That tryptophan is located in the motif **IGPxMGNxWIIYxYP** of group 1 proteins (see alignment position 460 in Fig. 3). The W residue is conserved in most BGLU proteins with the exception of BGLU11, 12, 13, 14, 22, 23 and 24 (Fig. 2). In those proteins, the motif is missing (Fig. 2).

The phenylalanine residues F193 and F200 (Zouhar et al. 2001) are located within the motif **NEPQTFTSFYSYGTGVFAPGRCS** in the BGLU1 protein (alignment position 250 in Fig. 2). Interestingly, not all BGLU proteins have conserved phenylalanines in those positions. For example, the motif **NEPQTFTSFSYG** is conserved only in the group 1 BGLU enzymes, whereas the phenylalanine of the motif **TGXFAFAPGRCS** is conserved in isozymes across several groups, such as in BGLU1, 2, 4, 6, 9, 11, 19, 15, 24 and 25 (Fig. 2). In other proteins of group 1, namely in BGLU3, 5, 7 and 8 that F residue is replaced by cysteine (Fig. 2), thus suggesting that those isoenzymes might have a different substrate specificity towards the aglycone part of the beta-glycoside.

The third phenylalanine residue F461 studied by (Zouhar et al. 2001) is located within the motif **DNFEWfaG-yTERYGI** of group 1 proteins (see alignment position 545 in Fig. 2). Again, that position is divergent among different BGLU enzymes, while BGLU1 and BGLU3 have a F residue, BGLU2 has a Y residue, whereas BGLU4 and BGLU6 have an S, BGLU7 has an T, and BGLU5 have an C residue. This is an additional indication of the highly divergent specificity that the different BGLU isoenzymes might have towards the beta-glycoside substrate.

Divergent sequences can be exploited in order generate specific probes for those genes. The alignments and detailed analysis of the sequences allowed us to define novel primer pairs that are specific for each *Zmbglu* isoform (Table S5). All primers were designed to span an intron, so that a cDNA template would generate an amplicon of different size as genomic DNA template. Those primer pairs need to be tested by various molecular breeders to validate them in a wide range of maize genotypes for marker applications and RT-PCR based expression assays of the *Zmbglu* genes.

Each *bglu* gene shows a particular combination of promoter elements that cannot predict the complex pattern of gene expression

Analysis of the promoter regions revealed that the *bglu* genes have a distinct set of *cis* elements and transcription factor binding sites (Table S1; Fig. S5). We found that

bglu7, 14, 17, 18, 20 and 22 do not contain a TATA box, the lack of such being characteristic of pseudogenes or non-expressed genes. However, using microarray we detected expression of those genes in maize silks (*bglu7* and *bglu20*), leaf in dark (*bglu14*) and light condition (*bglu17*), embryo (*bglu18*) and leaf under drought and well watered stress (*bglu22*) (Fig. 5). Although these genes do not have a classical TATA Box, they have many characteristic *cis* elements involved in multiple signaling pathways in plants like CGCGBOXAT in *bglu14*, 18, 20 and 22. The CCAAT box was found in *bglu14* and *bglu20*. It is a common DNA motif found immediately upstream from the most distal heat shock elements (HSE) of the promoter: they act cooperatively with HSE to increase the promoter activity (Rieping and Schoffl 1992). The *bglu17* gene has AMYBOX1 and AMYBOX2, common to most actively transcribed genes in plants. Also, the CAAT Box which is responsible for promoter activity, seed specificity, and temporal regulation of *legA* gene (storage proteins) was found in *bglu14*, 17, 18 and 20. The microarray results show that maybe all but *bglu10* are genes readily expressed in at least one tissue or environmental condition (Fig. 5). In addition to the known transcription binding sites, there might be additional *cis* elements and unknown factors that determine *bglu* gene transcription.

Initially, we had expected that the gene expression pattern could be roughly described by the presence of regulatory elements in the gene promotes. Contrary to our expectations, we found only a low overall correspondence between the dendrogram of gene expression (Fig. 5) and the dendrogram of *cis* regulatory elements (Fig. S5). This could imply that the *cis* elements promote gene expression not as one would naively predict from a simple linear additive model, but rather by a synergistic mechanism or a complicated interaction between different transcription factors. It could also imply that in addition to the known regulatory sequences (Table S1), an unknown set of *cis* elements determine the complex pattern of expression of the *bglu* gene family.

In summary, we propose that the combination of known and unknown regulatory sequences determine the complex pattern of expression of the *bglu* gene family. Thus, the presence of a given set of known *cis* elements alone does not allow making an accurate prediction of the expression pattern, making it necessary to measure mRNA levels and enzymatic activity in each experimental condition individually.

We did measure total BGLU activity in different tissues and conditions (Table 3 and 4) and conclude that the expression pattern of *bglu1* and *bglu2* (Fig. 5) alone cannot explain the activity observed in other maize tissues (e.g. silks). We therefore postulate that other BGLU paralogs

are indeed important in other tissues and under other circumstances.

bglu1 and *bglu2* show a distinct response under salt stress and mechanical damage

The microarray results and RT-PCR show that each one *bglu* gene has a particular expression pattern, tissue specificity and induced by abiotic conditions (Fig. 5). One of our original aims was to identify whether *bglu1* or *bglu2* are involved in salt stress and mechanical damage. The role of BGLU activity in abiotic stress could be related to the release of toxic molecules for defense or the activation of hormones for the regulation of growth. First we measured the β -glucosidase activity after seed germination. We then measured β -glucosidase activity in B73 maize coleoptiles under salt stress (300 mM NaCl) and mechanical damage (Fig. 6 and Table 3). We detected a significant increase in β -glucosidase activity after 30 min of salt stress. An increase of β -glucosidase in maize plants incubated with 100–200 mM NaCl has also been reported before (Dietz et al. 2000; Zorb et al. 2004). Since we could not discriminate if the increased activity in coleoptiles was mainly encoded by *bglu1* or by *bglu2*, we decided to analyze the mRNA levels using specific primers for *Zmbglu1* and *Zmbglu2* (Fig. S3). We detected a significant transient increase in *bglu1* but not of *bglu2* mRNA after 1 h of stress (Fig. 7). After 1 h of salt stress or mechanical damage the expression of *bglu2* rather decreased (Fig. 7). Since we observed an increase of activity but a decrease of *bglu2* transcript levels, we propose that *bglu2* is not involved in the stress response of coleoptiles. An alternative explanation is that mRNA and activity levels are not linked. This would be the case if the activity of BGLU enzymes requires post-translation modification to be activated. In arabidopsis, a rapid increase of activity has been reported under dehydration stress due to a fast polymerisation of the BGLU enzyme (Lee et al. 2006).

In order to better understand the differential expression between *bglu1* and *bglu2* we analyzed the promoter region of those genes. We found a specific *cis* element (WBOXNTCHN48) present in *bglu1* but absent in *bglu2* (Table S1). This element has been reported to be necessary in tobacco to link the elicitor induction by some transcription factors (Yamamoto et al. 2004). We propose that *bglu1* needs also elicitors to be highly induced after mechanical damage of maize coleoptiles. Other maize *bglu* genes with WBOXNTCHN48 *cis* element were: *bglu4*, 8, 13 and 24. Future work will concentrate in the evaluation of other *bglu* paralogs, to determine if they are also involved in insect attack or pathogen resistance in maize.

Comparison of the size of the *bglu* gene family in different plant species

We found that maize has less genes encoding for beta-glucosidase enzymes than other species. The arabidopsis genome contains 47 *bglu* genes (*Atbglu*) (Xu et al. 2004), whereas the rice genome has 34 *bglu* genes (*Osbglu*) (Opassiri et al. 2006). We found only 26 *bglu* genes in maize (*Zmbglu*) (Fig. 3 and Table 1). The difference between the number of *Atbglus* and *Zmbglus* could reflect the importance of glucosinolate metabolism that is typical of the *brassicacea* family to which arabidopsis belongs. In contrast to cabbage, cauliflower or broccoli, corn plants do not employ conjugates of inolate-glucose for pathogen defense. Since β -glucosidase enzymes are required for activating glucosinolates in the cabbage family, this may explain why rice and maize harbor less *bglu* gene copies, despite having a much larger genome. Arabidopsis has around 25 498 genes and a genome size of 125 MB (The Arabidopsis-Genome-Initiative 2000) whereas maize contains around 32 000 genes (at least) and a genome size of 2300 MB (Schnable et al. 2009). The maize group 1 enzymes are similar to arabidopsis AtBGLU13 and AtBGLU16, which have been classified into the AtBGLU group 2 of arabidopsis (Xu et al. 2004). All group 2 maize enzymes are most similar to arabidopsis AtBGLU11, and all group 3 maize enzymes are most similar to arabidopsis AtBGLU44 (Table 2). AtBGLU11 belongs to group 1 of arabidopsis and AtBGLU44 belongs to arabidopsis group 9 (Xu et al. 2004). All this indicates that in comparison to arabidopsis, maize contains less groups of BGLU enzymes (less divergent sequences), but that some groups of ZmBGLU proteins seem to be more numerous, which suggest a higher importance for maize defense or development.

The range of substrate specificity and the degree of enzyme promiscuity of each isoform will need to be analyzed in future experiments. The list of BGLU orthologs in other plant species (Table 2) can help to postulate possible physiological roles for each member of the maize family, since arabidopsis mutants can be more readily obtained and grown for phenotypic characterization.

Conclusions

Use of molecular information for plant breeding in the postgenomic era

Comparison of the *bglu* gene expression across tissues, genotypes and environments (Fig. 5) allowed us to make an estimate on the magnitude of the environmental and genetic variances. We conclude that the variance due to different genetic backgrounds (even from different

heterotic groups) is relatively small in comparison to the variance due to the environment or the tissue analyzed. In other words, it matters more if the plants are grown in the greenhouse or the field, if they are grown in México, USA or Europe, than if one chooses the one or the other genotype. It matters more which tissue is analyzed (e.g. roots or leaves) than the environmental conditions (e.g. light–dark or drought (SS) and well watered (WW) conditions) (Fig. 5).

This is bad news for gene-expression experiments in which the same genotypes are grown under different field conditions in different countries. Results from one lab seldom can be replicated in another environment. It is also bad -but old- news for plant improvement. Breeders have to work with a relatively small amount of genetic variance in order to obtain yield gains. The large magnitude of environmental variance is the biggest problem for any breeding program, particularly for those trying to improve stress tolerance. The gene expression variance due to the environment is larger than the variance due to different genetic backgrounds (Fig. 5). However, the gene expression variance in different tissues is much larger than the sum of both environmental and genetic variance.

The old knowledge about environmental variance can be translated to a practical advice like: in order to obtain higher yields, one does not need to spend much effort in genetic selection, just add more fertilizer and pesticide to any of your genotypes. In the same context, the knowledge about tissue variance can be translated to something similar as: in order to increase the disease resistance of a susceptible tissue like the ears, try obtaining the same gene expression pattern of non-susceptible tissues like stems or roots.

For a biotechnological application it can be said that if scientists need more genetic diversity for disease resistance, the potential exists for a much greater range of *bglu* gene expression using endogenous genetic elements that define tissue specific expression levels (Table S1). Molecular maize breeders would require rearranging the endogenous *cis* elements that could contribute to the desired phenotypic trait in a given tissue. It is not a trivial observation that some tissues are more susceptible to pathogen attack than other tissues. This could be reflecting tissue specific expression of some resistance proteins such as BGLU. Molecular breeding approaches can therefore benefit from the identification of all gene family members responsible for a given trait. The substrate specificity of relevant defense enzymes is also important. In addition, the identification/validation of relevant *cis* sequences that regulates gene expression in different tissues and conditions (GEN x ENV x TIS interactions) can provide valuable information for better tapping the challenges of plant breeding, which aims to produce more food with the available resources and endogenous genes for each crop.

Overall, we postulate that the loci shown in Fig. 8 correspond to functional enzymes with BGLU activity in the maize genome and therefore could be relevant for molecular breeding approaches (e.g. for abiotic and biotic stress, disease resistance, etc.).

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